



CS & IT ENGINEERING



Algorithms

Dynamic Programming

Lecture No.- 04



By- Aditya sir

Recap of Previous Lecture



Topic

Topic

BF

FW

0/1 Knap



Topics to be Covered



Topic

Topic

Topic

0/1 Knap
LCS



About Aditya Jain sir

1. Appeared for GATE during BTech and secured AIR 60 in GATE in very first attempt - City topper
2. Represented college as the first Google DSC Ambassador.
3. The only student from the batch to secure an internship at Amazon. (9+ CGPA)
4. Had offer from IIT Bombay and IISc Bangalore to join the Masters program
5. Joined IIT Bombay for my 2 year Masters program, specialization in Data Science
6. Published multiple research papers in well known conferences along with the team
7. Received the prestigious excellence in Research award from IIT Bombay for my Masters thesis
8. Completed my Masters with an overall GPA of 9.36/10
9. Joined Dream11 as a Data Scientist
10. Have mentored 12,000+ students & working professions in field of Data Science and Analytics
11. Have been mentoring & teaching GATE aspirants to secure a great rank in limited time
12. Have got around 27.5K followers on LinkedIn where I share my insights and guide students and professionals.



Telegram Link for Aditya Jain sir: https://t.me/AdityaSir_PW

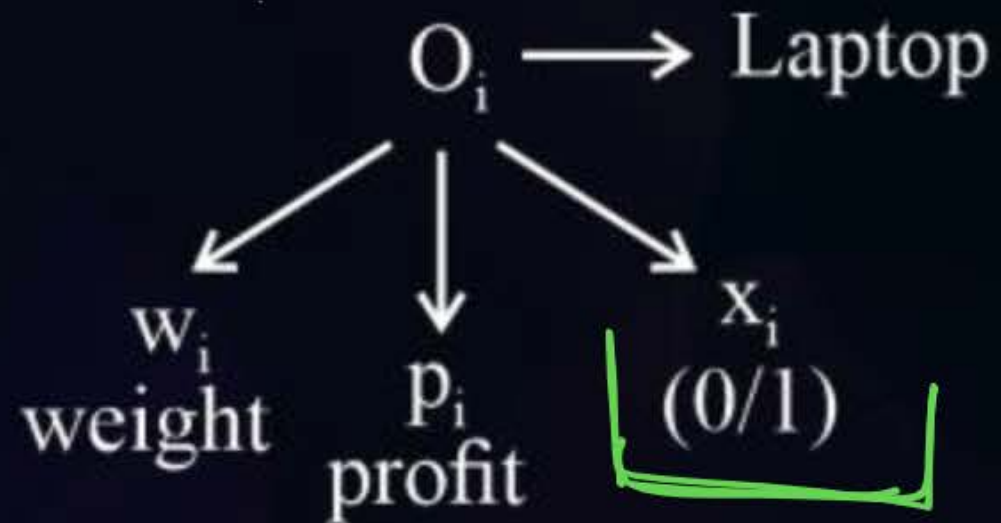
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Topic : Dynamic Programming

0/1 Knapsack (Binary Knapsack):

- Knapsack Capacity : M
- no. of objects $\rightarrow n$





Topic : Dynamic Programming



Objective Function:

$$\max \sum_{i=1}^n p_i * x_i$$

Such that:

$$\sum w_i * x_i \leq M$$

$$x_i = \{0, 1\}$$



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Dynamic Programming Approach:

- Let $01 \text{ Knap}(n, M)$ represent ^{max} profit with n -objects and Knapsack of Capacity M .



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Recurrence:

$$01 \text{ Knap}(n, M) = 01 \text{ Knap}(n-1, M), \underline{w_n > M} \rightarrow \underline{\text{Exclude } O_n \text{ (no option)}}$$

$$01 \text{ Knap}(n, M) = \underline{\max} \left\{ \begin{array}{l} 01 \text{ Knap}(n-1, M) \\ \underline{01 \text{ Knap}(n-1, M - w_n)} + \underline{p_n} \end{array} \right\}, \quad \underline{w_n \leq M}$$

Initialization,

$$\underline{01 \text{ Knap}(n, M) = 0, \text{ if } n = 0 \text{ or } M = 0}$$



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Example:

To understand if there are overlapping Sub-problem or not?

$n = 4$, $M = 3$

	01	02	03	04
P_i profits	10	20	30	40
W_i weights	1	1	1	1

Top-Down Tree

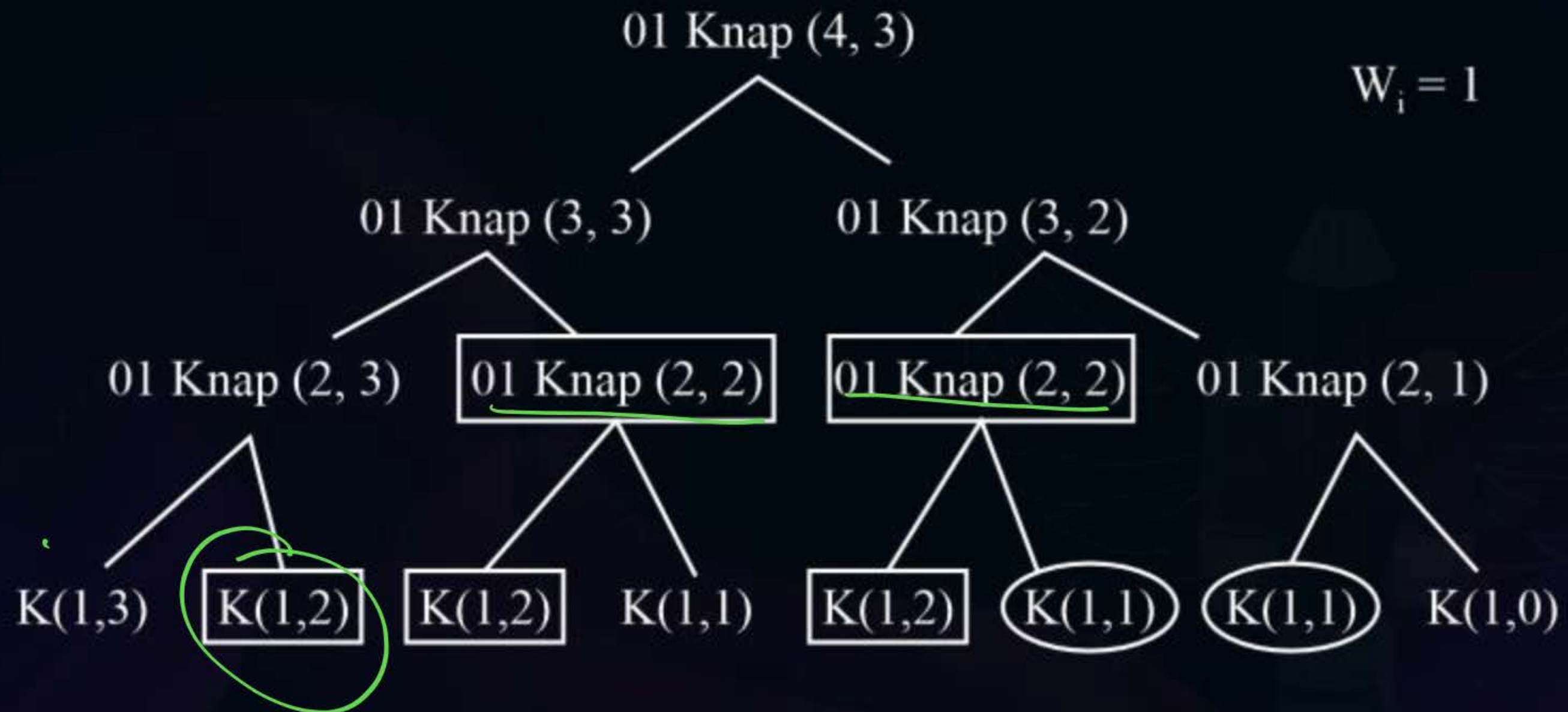
0/1 Knap (4, 3)



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Top-Down (Recursive):





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Bottom-up Approach (Tabulation):

Example:

$$\underline{n = 4}, \underline{M = 8}$$

	01	02	03	04
P_i	1	2	3	6
W_i	2	3	4	5

Tabulation:

Let $X[0 \dots n, 0 \dots M]$ be an array of which $X[n, M] \Rightarrow$ final answer (max profit)



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M

P_i	W_i
1	2
2	3
5	4
6	5

$n = 4$
 $m = 8$

	0	1	2	3	4	5	6	7	8
0	0	0	0	0	0	0	0	0	0
1	0	0	1	1	1	1	1	1	1
2	0	0	1	2	2	3	3	3	3
3	0	0	1	2	5	5	6	7	7
4	0	0	1	2	5	6	6	7	8

Knap(4, 8)



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$$X[i, j] = \max\{X[i-1, j], X[i-1, j-W_i] + P_i\}$$

$$X[2, 3] = \max\{X[1, 3], X[1, 3-3] + 2\}$$

$$X[2, 3] = \max\{1, 0+2\} = \max[1, 2] = 2$$

$X[2, 5]$ = max profit we can have with considering only 1st two objects and
knapsack of capacity 5.



Topic : 0/1 Knapsack Code

```
Algo 01Knap(M, n, W, P) {  
    X[0...n, 0...M]  
    for (i = 0; i <= n; i++) {  
        for (j = 0; j <= M; j++) {  
            if (i == 0 or j == 0) {  
                X[i, j] = 0  
            } else if (W[i] ≤ j) {  
                X[i, j] = max(X[i-1, j], X[i-1, j-W[i]] + P[i])  
            } else {  
                X[i, j] = X[i-1, j]  
            }  
        }  
    }  
    return X[n, M]  
}
```

0	0	0	0	0	0
0					
0					
0					
0					
0					



Topic : 0/1 Knapsack Code

X	0	1	2	.	.	.	M
0							
.							
.							
.							
.							
n							



Topic : 0/1 Knapsack Code

#Q. $n = 3, M = 6$

0/1

Solve

Hw $\rightarrow$ Table

	01	02	03
P_i	1	2	5
W_i	2	3	4

Diagram showing a bracket under the W_i row (values 2, 3, 4) with arrows pointing down to the numbers 3 and 6, indicating a cumulative weight calculation.



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Complexity Analysis of 0/1 Knapsack:

1. Time Complexity:

- $O(n * M)$

2. Space Complexity:

- $O((n+1) * (M+1)) = O(n * M)$



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Longest Common Sub-Sequence (LCS):

- Based on Strings
- A sequence/group of one or more characters.



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String:

1) Substring:

- A contiguous sequence of 1 or more characters from the given string.



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String:

2) Subsequence:

- A sequence of 1 or more characters from the given string, that may or may not be contiguous, but their relative order is maintained.



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Example:

- String: "ABC" $\rightarrow$ length = 3
- Sub-strings: A, B, C $\rightarrow$ 3

AB, BC $\rightarrow$ 2

ABC $\rightarrow$ 1

No. of substrings = $3 + 2 + 1 = 6$



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Important Result:

Given : $S[1, \dots, n]$

#Q. What are the total number of sub-strings of at-least length 1 (non-empty sub-strings)?

$$\left[\frac{n(n+1)}{2} \right]$$



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Sub-Sequence:

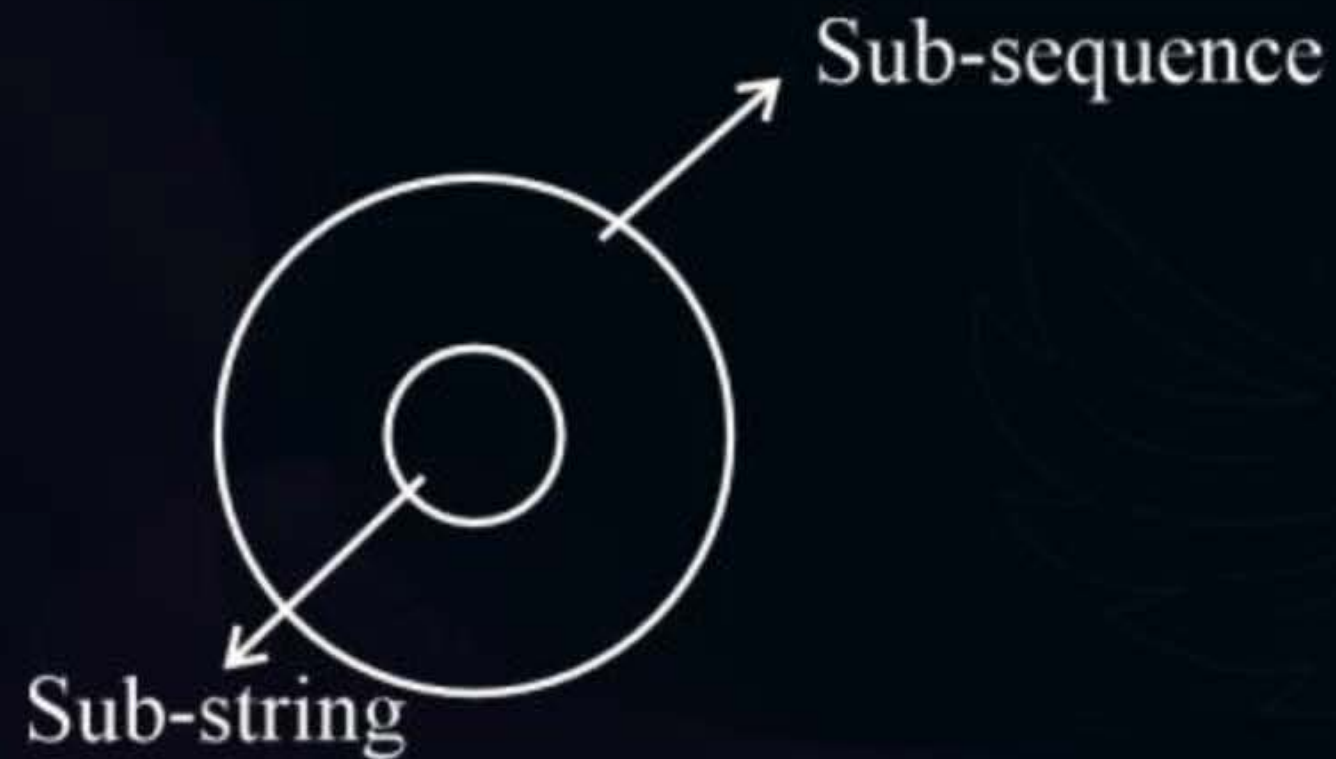
- String = "ABC"
- Sub-Sequence = A, B, C, AB, BC, AC, ABC → Valid sub-sequence
- CA, CB → relative order not maintained, not a sub-sequence.



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Important:

- Every substring is also a sub-sequence, but a Sub-sequence may or may not be a substring.



[MSQ]

#Q. Given a string $S = \text{"CABDABBBCD"}$.
Which of the following are valid sub-sequences?

- A** BBD ✓
- B** BAC ✓
- C** CDCA ✗
- D** CADC ✓
- E** ADABD ✓

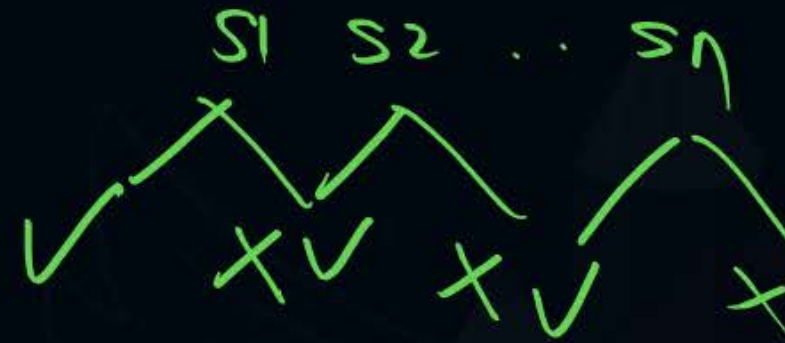


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Longest Common Sub-Sequence:

#Q. Given a string of length 'n' characters, the number of sub-sequences possible are?

$$2^n$$



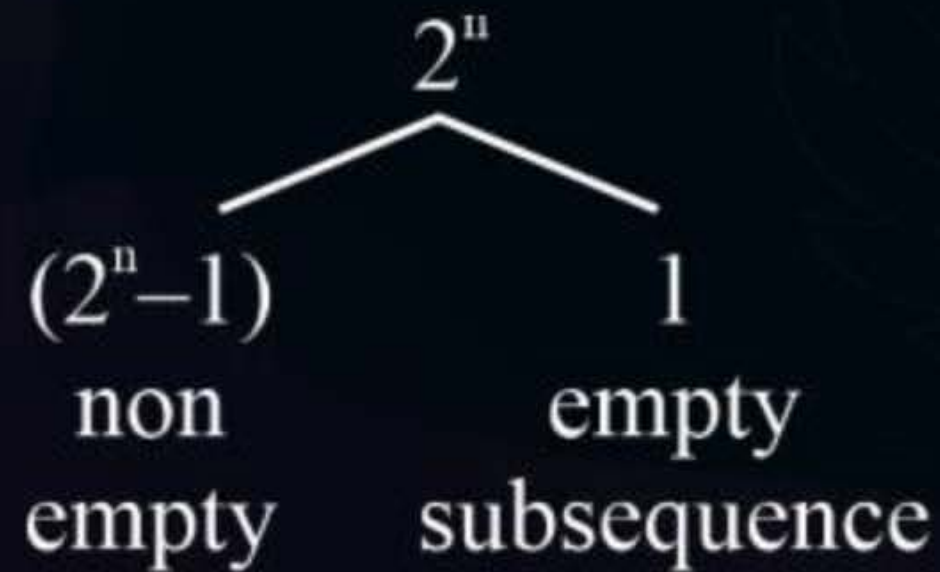


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Longest Common Sub-Sequence:



$\phi \rightarrow$ empty sub-sequence





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Brute Force/Enumeration:

All Subsequence $\rightarrow A$

All Subsequence $\rightarrow B$

- $O(2^n)$
- Hence, we need DP based approach.



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Longest Common Subsequence (DP based approach):

Common Subsequence:

- Given two strings X and Y of length n and m characters, a subsequence that is common to both X and Y is known as a common sub-sequence.



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Example:

#Q. X = "ABCD", Y = "BDC"

Which sub-sequences are common to both?

1) A ✗

2) B ✓

3) BC ✓

4) BD ✓

5) BCD ✗

6) DC ✗

7) BDC ✗



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LCS Problem Statement:

#Q. Given strings 'A' and 'B' of length (Characters) 'n' & 'M' respectively. The LCS problem is to determine the subsequence that is of the longest length among all the subsequences that are common to both 'A' & 'B'?



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Example:

$S1 = \text{"ABCB DAB"}$

$S2 = \text{"BDCABA"}$

Longest LCS = 4



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Common Subsequences:

AB $\rightarrow$ 2

BCA $\rightarrow$ 3

BCB $\rightarrow$ 3

BCBA $\rightarrow$ 4

BCAB $\rightarrow$ 4

BD AB

- Multiple longest common subsequences are possible.
- But LCS length is unique $\rightarrow$ 4



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DP based solution to LCS:

- Let 'i' and 'j' be indices in the strings 'X' and 'Y' respectively, such that.

$$X = [x_1, x_2, x_3, \dots, x_n]$$

$$Y = [y_1, y_2, y_3, \dots, y_m]$$



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- Let $L(i, j)$ represent the length of the common sub-sequence of the strings X and Y .

$$L(i, j) = 1 + L(i-1, j-1); \text{ if } \underline{X[i] == Y[j]}$$
$$= \max\{L(i-1, j), L(i, j-1)\}; \text{ if } \underline{\underline{X[i] \neq Y[j]}}$$

Initialization

$$\frac{i=0}{j=0}$$





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Example:

Case1: $X[i] = Y[j]$

$X = \text{"ABBCDCCABA"}$

$Y = \text{"CBDADCBDAA"}$

It



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Example:

Case2: $X[i] \neq Y[j]$

$X = \text{"ABCDDBCCADB"}$

$Y = \text{"BDBC AAABDCD"}$

$$\max\{\underbrace{L(i, j-1)}, \underbrace{L(i-1, j)}\}$$

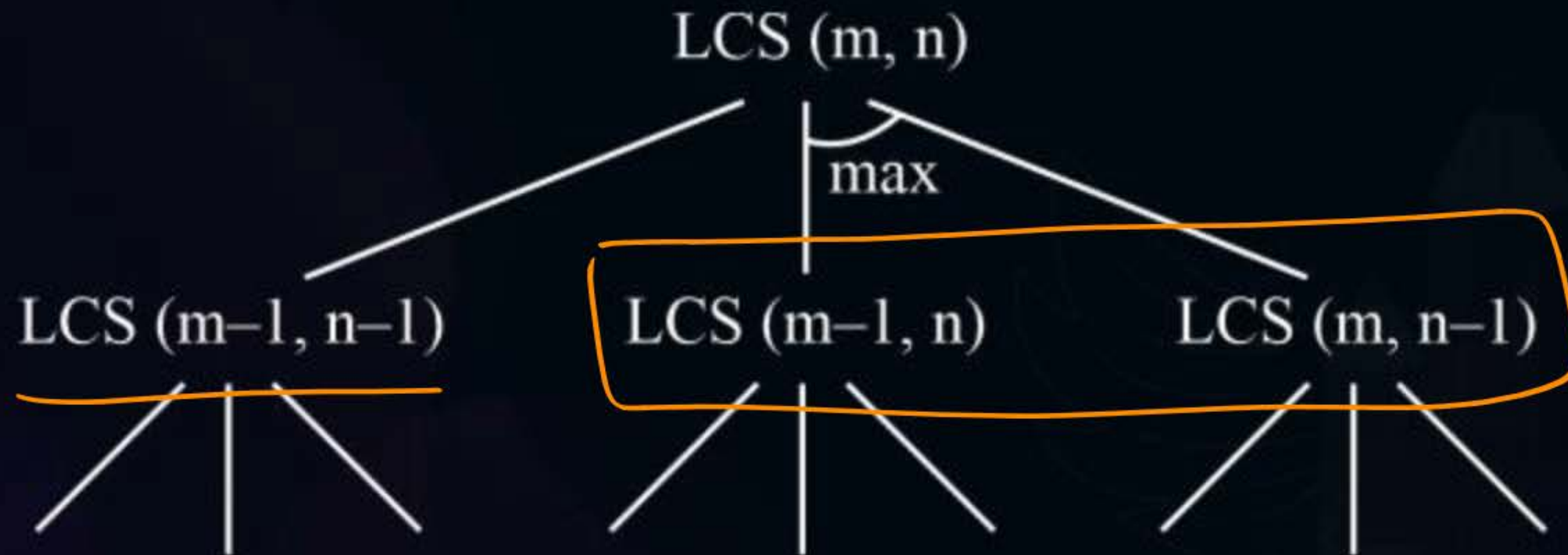


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Overlapping Subproblems:

$$|X| = n, |Y| = m$$

$$n=4, m=5$$

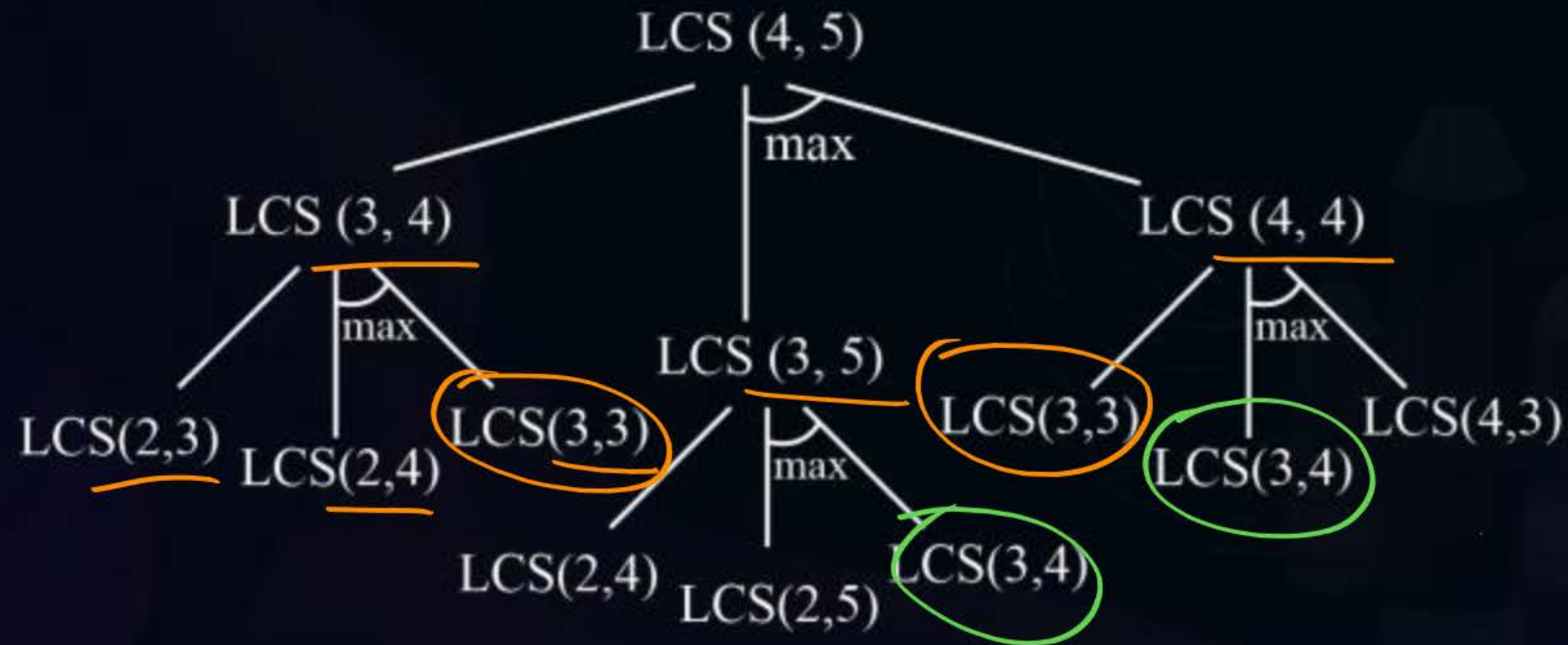




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Overlapping Subproblems:

$$|X| = 4, |Y| = 5$$





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Example:

Top-Down (Memoization/Recurrence):

X = "ABBAB"

Y = "ACBAB"

LCS : "ABAB"

Length = 4

$x = \text{ABBAR}$
 $y = \text{ACBAB}$



BABBA
 $\Downarrow$

$\text{rev} \Rightarrow \underline{\underline{\text{ABABA}}}$

$\text{LCS}(\text{AB}, \text{A})$

$\text{LCS}(\text{AB}, "")$

$\text{A} \quad \text{LCS}(\text{A}, \text{A}) \rightarrow 1 + \text{LCS}("", "") = 0$

$\text{A} \quad \text{LCS}(\text{A}, \text{A}) \rightarrow 1 + \text{LCS}("", "") = 0$

$\text{LCS}(\text{A}, \text{AC})$

$\text{LCS}("", \text{AC})$



2 mins Summary



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0/1 Knapsack

LCS



THANK - YOU